

Sharp Wave-Ripple Complexes (SWRs) and Their Mechanisms

Lea Friedemann, Shokoofeh Hosseini, Max Nowaczyk

WiSe 24/25

1 Introduction

Sharp wave-ripple complexes (SWRs) are network phenomena that occur in the hippocampus, particularly within the CA3 region of the mammalian brain. They are characterized by brief events of elevated and synchronized neuronal activity lasting about 50 to 100 milliseconds (Evangelista et al., 2020). SWRs primarily manifest during periods of slow-wave sleep or quiet wakefulness and play a crucial role in memory consolidation, especially in reinforcing episodic memory.

During these events, sequences of neuronal activity—associated with learned experiences—are replayed, which is believed to strengthen the memory traces. Experimental evidence suggests that blocking SWRs impairs spatial memory performance, highlighting their importance for learning.

2 Dynamics and Mechanisms of SWRs

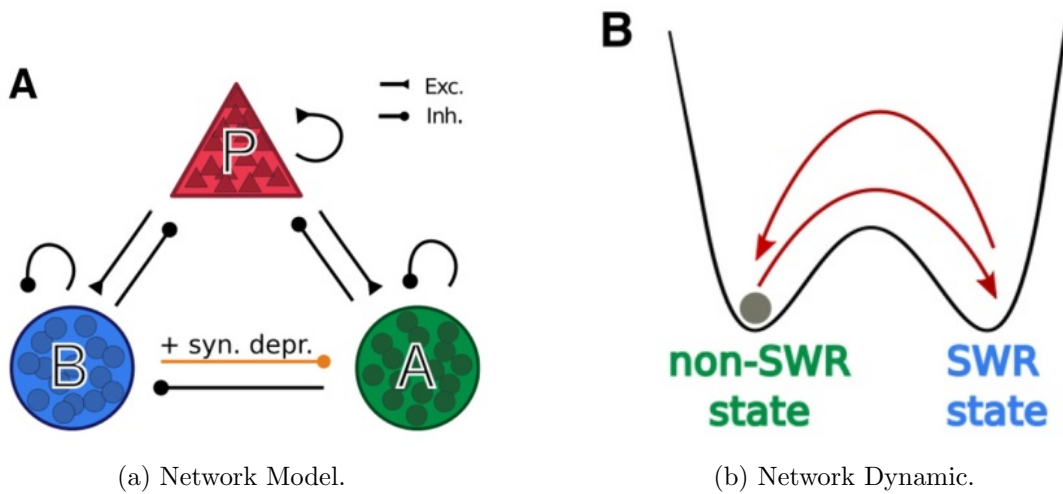


Figure 1: Comparison of network model and dynamics. Adapted from (Evangelista et al., 2020)

Neuronal Activity Patterns

Pyramidal cells increase their firing rate during SWRs. Parvalbumin-positive (PV⁺) basket cells show a threefold increase in firing during these events. Another class of interneurons, referred to as anti-SWR cells, becomes less active during SWRs, facilitating the disinhibition of pyramidal cells.

Triggering and Termination

SWRs can be triggered by the activation of pyramidal cells or PV⁺ basket cells. Synaptic depression in PV⁺ basket cells is thought to modulate the termination of SWR events, making them transient and preventing prolonged activation.

Disinhibition Mechanism

The initiation of SWRs is explained by a disinhibitory mechanism, where increased activity of PV⁺ basket cells suppresses anti-SWR cells, leading to a release of inhibition on pyramidal cells. This interaction results in a high-activity SWR state. The network then transitions back to a low-activity, non-SWR state when anti-SWR cells regain control, inhibiting pyramidal cells.

3 Methods

Network Setup

The network consists of Pyramidal cells (P): Excitatory neurons, with a total of 8200 cells. PV⁺ Basket cells (B): Inhibitory interneurons, totaling 135 cells. Anti-SWR cells (A): Another inhibitory interneuron population, with 50 cells. These cells are recurrently connected based on known CA3 hippocampal connectivity. Synaptic connection probabilities are modeled according to experimental data.

Neuron Model

The neurons are simulated using a leaky integrate-and-fire (LIF) with conductance-based synapses model. The subthreshold membrane potential is described by the following equation:

$$C \frac{dV_i}{dt} = g_L(V_{\text{rest}} - V_i) - [g_P(t)(V_i - E_P^{\text{rev}}) + g_A(t)(V_i - E_A^{\text{rev}}) + g_B(t)(V_i - E_B^{\text{rev}})] + I_{\text{ext}} \quad (1)$$

Here, $g_P(t)$, $g_A(t)$, and $g_B(t)$ are synaptic conductance for pyramidal, anti-SWR, and PV⁺ basket cells, respectively.

The total synaptic conductances of synapse i are governed by the following equations:

$$\begin{aligned}
\frac{dg_i^P}{dt} &= -\frac{g_i^P}{\tau_{\text{syn}}^P} + \sum_{f,j} \delta(t - t_j^{(f)} - \tau_{IP}) g_{ij}^{IP}, \\
\frac{dg_i^B}{dt} &= -\frac{g_i^B}{\tau_{\text{syn}}^B} + \sum_{f,j} \delta(t - t_j^{(f)} - \tau_{IB}) g_{ij}^{IB}, \\
\frac{dg_i^A}{dt} &= -\frac{g_i^A}{\tau_{\text{syn}}^A} + \sum_{f,j} \delta(t - t_j^{(f)} - \tau_{IA}) g_{ij}^{IA},
\end{aligned} \tag{2}$$

$$I \in \{A, B, P\}$$

Synaptic plasticity

In order to allow for the generation of SWR events a synaptic short-term depression mechanism was introduced. Synaptic increases in the B→A synapse are scaled by a synaptic efficacy variable, e_{AB} .

The synapse evolves over time according to the following equation:

$$\frac{de_{AB}}{dt} = \frac{1 - e_{AB}}{\tau_D} - \sum \delta(t - t_{\text{spike}}) e_{AB} \eta_D \tag{3}$$

The synaptic efficacy is modeled separately for each individual B→A synapse. Default parameters are $\tau_D = 250$ ms and $\eta_D = 0.18$.

State Analysis

The network is bistable, meaning it can switch between two stable states:

- **Non-SWR State:** High activity in A cells, controlling P and B cells. Low activity in P and B cells (2 spikes/s and 1 spike/s, respectively).
- **SWR State:** Increased activity in P and B cells (43 spikes/s and 90 spikes/s, respectively). A cells become mostly inactive (1 spike/s).

Simulation Procedure

State switching

To observe the switch between SWR and non-SWR state, de- and hyperpolarizing currents were injected into the network. The simulation was started with just constant background current of 200 pA and synaptic efficacy e_{AB} clamped at 0.5. In order to switch the network state to the SWR state, a depolarizing current was injected into 60% of the cells in population P. The current intensity was uniformly distributed between 0 and 600 pA.

The network was then left to evolve for 1000 ms. To switch the network back the non-SWR state, a hyperpolarizing current was injected, again into 60% of the cells in population P. The current intensity was uniformly distributed between 0 and 500 pA.

Spontaneous SWR Events

The network starts in a non-SWR state, with e_{AB} initialized at 0.5 and not clamped. All populations of cells receive a constant background current of 200 pA. The network is then left to evolve over time and SWR events occur spontaneously.

Quantification of SWR Events

Detection criteria for SWR events include:

- Increased firing rates in the P and B populations.
- Decreased firing rates in the A population.

We conducted evaluation metrics such as the duration of SWR events, Inter-event intervals (IEI) between consecutive SWRs, Full Width at Half Maximum (FWHM), Peak frequency in the power spectrum, and Population firing rates in both SWR and non-SWR states.

Simulation and Data Analysis Tools

The simulations are performed using the Brian simulator for spiking neural networks. Data analysis, including population rates and event quantification, is conducted in Python with Gaussian smoothing for instantaneous rates.

4 Results

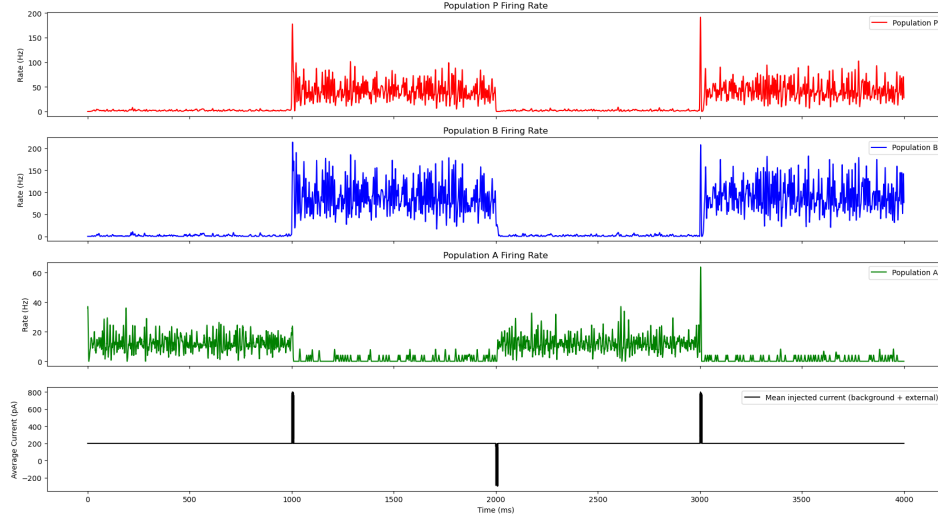


Figure 2: Simulation results for the synaptic efficacy clamped at 0.5. A depolarizing current was applied at 1000 ms and 3000 ms to induce the SWR state and a depolarizing current at 2000 ms to restore the non-SWR state.

First, we verified the implementation's bistability by clamping e_{AB} at 0.5, where no spontaneous state transitions occur, indicating both states are stable. Delivering an appropriate current, however, could induce a state switch to the other stable state (Figure 2).

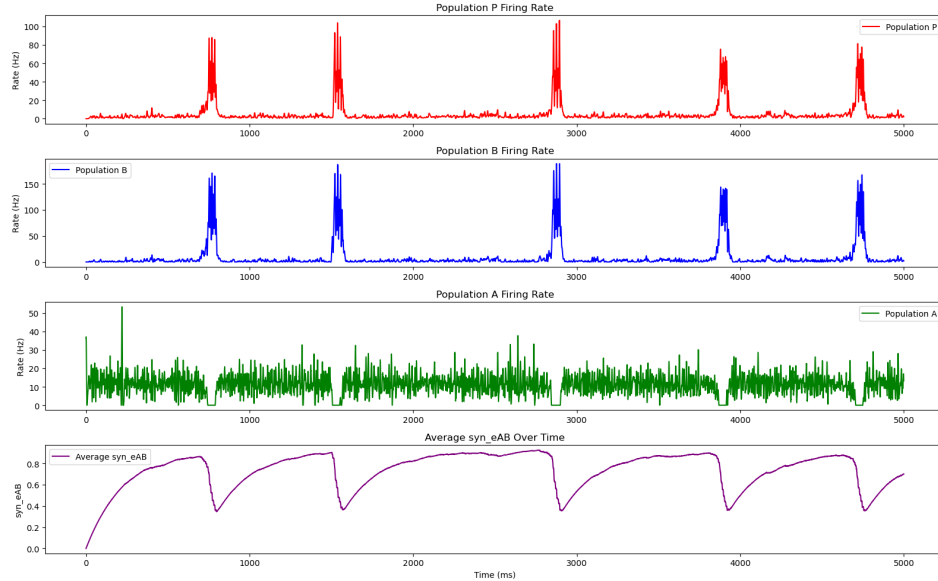


Figure 3: If the synaptic efficacy is unclamped, SWRs occur and are terminated spontaneously.

If e_{AB} is unclamped, spontaneous SWRs occur when the efficacy is close to 0.8, followed by a steep decrease in efficacy due to the synaptic depression mechanism and thus termination of the event (Figure 3).

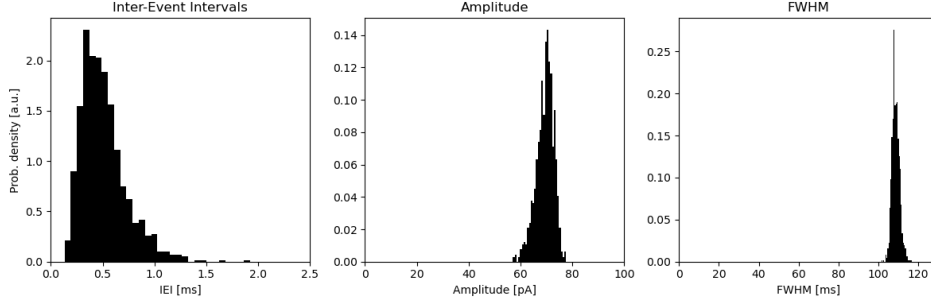


Figure 4: Properties of spontaneous SWRs

For the spontaneous SWRs measured over 10 minutes, the observed properties like IEI, event amplitude and FWHM were in the same range as reported by Evangelista et al. (2020) (Figure 4). Furthermore, we could also reproduce the strong and highly significant correlation between the amplitude of an SWR and the previous IEI (Figure 5).

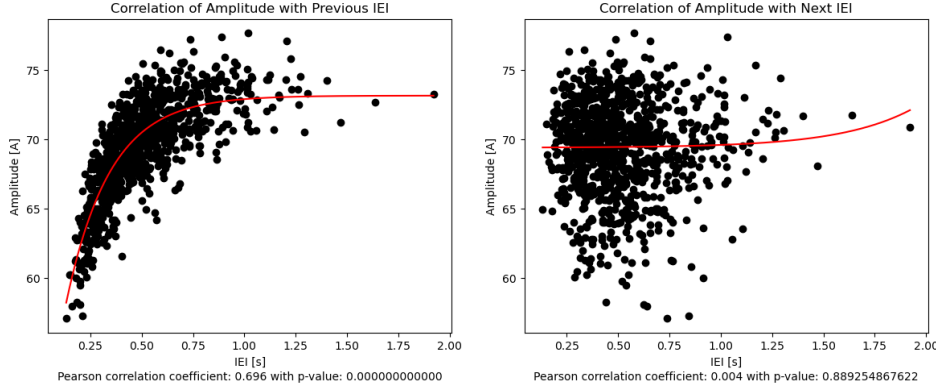


Figure 5: Correlation of the event amplitude with the previous (strong correlation) and next (weak correlation) IEI

5 Discussion

To summarize, we could successfully replicate several findings of X, such as the network's bistability, as well as the properties and statistics of the SWRs. The proposed mechanism accurately captures some properties of experimentally observed SWRs, such as the incidence, which is observed to be 1-2 in experiments, or the correlation between event amplitude and the previous Inter-Event Interval.

While SWRs can also be simulated using different methods, like learning of appropriate weights (Ecker et al., 2022) that do not require a third population, disinhibition is a promising mechanism as it is widespread in interneurons of the cortex and could explain the paradoxical involvement of PV+ basket cells in the SWRs.

However, to prove this proposal, at least one neuron population corresponding to the anti-SWR cells has to be identified. While the identity of those cells in CA3 is unclear,

cholecystokinin-expressing (CCK) basket cells in CA1 could be a promising candidate (Dudok et al., 2021). They show reciprocal inhibitory connections with PV+ BC. They show CCK BCs are strongly inhibited by PV+ BCs, and show decreased activity during SWR. Further research is necessary to examine the role CCK BC could play in SWR and if there is a similar group of interneurons CA3.

References

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